



Universidad Carlos III de Madrid

Aerospace Engineering

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# Computer Laboratory Report 1

Particle Connected to a Spool

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Mechanics Applied to Aerospace Engineering

Laboratory Group LA

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## 1 Introduction

In this laboratory experiment, the planar dynamics of a particle  $P$  of mass  $m$  connected by a thin, massless, inextensible string of total length  $l$  to a cylindrical spool of radius  $a$  centered at the origin  $O$  is analyzed. The spool rotates at a constant angular velocity  $\omega$  in the counterclockwise direction. The instantaneous contact point between the string and the spool is denoted by  $B$ , and the system geometry is parametrized by the angle  $\phi$  between the horizontal axis and the radius vector  $OB$ .

The main objective is to investigate the dynamic behavior of the particle under different spool angular velocities and initial velocities, examining trajectory shapes, unwound length evolution, string tension, and mechanical energy conservation. Five representative cases are analyzed and integrated numerically using `ode45`.

## 2 Methodology

### 2.1 Study of the Degrees of freedom

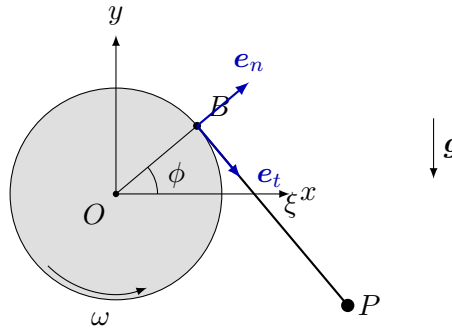


Figure 1: Geometry of the problem and local basis  $\{e_n, e_t\}$  attached to the point of tangency  $B$ .

Let  $e_n = (\cos \phi, \sin \phi)$  be the unit vector along  $OB$  and  $e_t = (\sin \phi, -\cos \phi)$  the unit vector tangent to the spool at  $B$ , oriented along the free segment from  $B$  to  $P$  (Fig. 1). Since  $\dot{\phi}$  is the rotation rate of both vectors,

$$\dot{e}_n = -\dot{\phi} e_t, \quad \dot{e}_t = \dot{\phi} e_n. \quad (1)$$

**Degrees of freedom.** The particle  $P$  is free on a plane, so it has two coordinates to move along. The string attached to the particle imposes a constraint: the sum between length of the wound arc (let the attachment point be  $A$ )  $AB$  and the length of the segment  $BP$  must equal  $l$ , and  $l = AB + BP = \text{constant}$ . The point of tangency can vary, but the total length of the string cannot. Hence,  $2 - 1 = 1$  configuration degree of freedom, which can be parametrised with  $\phi$ . If  $\alpha(t) = \alpha_0 + \omega t$  is the angular position of the attachment point on the spool, the arc wound on the spool (clockwise from the attachment point to  $B$ ) is  $a[\alpha(t) - \phi]$ , where  $[\alpha(t) - \phi]$  is the angular difference between the attachment point and  $B$ , and the inextensibility  $l = a[\alpha(t) - \phi] + \xi$ , where  $\xi$  is the variable length of the segment  $BP$ , gives, using  $\xi(0) = \xi_0$  and  $\phi(0) = 0$ ,

$$\xi(t, \phi) = \xi_0 + a\phi - a\omega t. \quad (2)$$

Therefore the position of  $P$ ,

$$\mathbf{r}_P = a \mathbf{e}_n + \xi \mathbf{e}_t = \begin{pmatrix} a \cos \phi + \xi \sin \phi \\ a \sin \phi - \xi \cos \phi \end{pmatrix}, \quad (3)$$

is completely determined by  $\phi$  and  $t$ : the system has **one configuration degree of freedom**, parametrised by  $\phi$ . Geometrically,  $P$  lies on the involute of the spool circle; for  $\omega \neq 0$  the involute rotates rigidly with the spool (its starting angle is  $\omega t - \xi_0/a$ ), so the constraint depends explicitly on time. For  $\omega = 0$  the constraint is time independent.

**Velocity and acceleration.** Using Eq. (1) and  $\dot{\xi} = a(\dot{\phi} - \omega)$ ,

$$\mathbf{v}_P = (\dot{\xi} - a\dot{\phi}) \mathbf{e}_t + \xi \dot{\phi} \mathbf{e}_n = \xi \dot{\phi} \mathbf{e}_n - a\omega \mathbf{e}_t, \quad (4)$$

$$\mathbf{a}_P = (\xi \ddot{\phi} + a\dot{\phi}^2 - 2a\omega \dot{\phi}) \mathbf{e}_n - \xi \dot{\phi}^2 \mathbf{e}_t. \quad (5)$$

Equation (4) shows that the velocity component along the string is fixed by the spool,  $\mathbf{v}_P \cdot \mathbf{e}_t = -a\omega$  (the string cannot stretch), and that the component normal to the string is  $\xi\dot{\phi}$ . The initial condition therefore satisfies  $\dot{x}_P(0) = \xi_0\dot{\phi}(0)$ , i.e.  $\dot{\phi}(0) = \dot{x}_P(0)/\xi_0$ , while  $\dot{y}_P(0) = a\omega$  is not free but imposed by the constraint.

## 2.2 Forces and equation of motion

Only two forces act on  $P$ : its weight and the tension  $T$  of the string, which pulls  $P$  towards  $B$ :

$$\mathbf{F} = -mg\mathbf{j} - T\mathbf{e}_t, \quad T \geq 0, \quad -mg\mathbf{j} = mg\cos\phi\mathbf{e}_t - mg\sin\phi\mathbf{e}_n. \quad (6)$$

Writing Newton's second law  $m\mathbf{a}_P = \mathbf{F}$  in the directions  $\mathbf{e}_n$  and  $\mathbf{e}_t$ :

$$\xi\ddot{\phi} + a\dot{\phi}^2 - 2a\omega\dot{\phi} = -g\sin\phi, \quad (7)$$

$$T = m(g\cos\phi + \xi\dot{\phi}^2). \quad (8)$$

The tension does not appear in the first equation, because it has no component along  $\mathbf{e}_n$ . So the first equation alone gives the motion:

$$\ddot{\phi} = -\frac{g\sin\phi + a\dot{\phi}^2 - 2a\omega\dot{\phi}}{\xi_0 + a\phi - a\omega t}, \quad (9)$$

and once  $\phi$  and  $\dot{\phi}$  are determined, Eq. (8) provides the string tension. The physical significance of each term in Eq. (9) can be interpreted as follows:

- If the spool has negligible radius ( $a \rightarrow 0$ ), the formulation reduces to a standard simple pendulum of constant length  $\xi_0$ :  $\ddot{\phi} = -(g/\xi_0)\sin\phi$ .
- For  $a > 0$ , the pendulum length varies continuously as the string wraps or unwinds, introducing the nonlinear centrifugal coupling term  $a\dot{\phi}^2$ .
- The term  $-2a\omega\dot{\phi}$  acts as a Coriolis-like acceleration caused by the rotation of the spool.
- When  $\xi \rightarrow 0$  (the unwound string is completely wrapped), the denominator vanishes: the particle reaches the spool surface and the taut-involute model terminates.

Two direct conclusions follow. First, if  $\phi(0) = 0$  and  $\dot{\phi}(0) = 0$ , the right-hand side of Eq. (9) vanishes identically and  $\phi$  remains zero: the particle hangs vertically downward. Second, because a flexible string supports tensile but not compressive stress, the condition  $T \geq 0$  requires  $\xi\dot{\phi}^2 \geq -g\cos\phi$ . This condition can only be violated if  $\cos\phi < 0$ , which occurs when the string points above the horizontal through  $B$  and the particle moves slowly.

## 2.3 Mechanical energy

From Eqs. (3) and (4), with  $|\mathbf{v}_P|^2 = \xi^2\dot{\phi}^2 + a^2\omega^2$  and  $y_P = a\sin\phi - \xi\cos\phi$ ,

$$E = \frac{1}{2}m(\xi^2\dot{\phi}^2 + a^2\omega^2) + mg(a\sin\phi - \xi\cos\phi). \quad (10)$$

Weight is conservative and is included in  $E$ , so  $\dot{E}$  equals the power of the tension,  $-T\mathbf{e}_t \cdot \mathbf{v}_P$ , that is

$$\frac{dE}{dt} = a\omega T. \quad (11)$$

When  $\omega = 0$ , the string tension is perpendicular to the velocity vector, performing zero mechanical work; consequently, mechanical energy is strictly conserved. When  $\omega > 0$ , the spool performs work on the particle at the positive rate  $dE/dt = a\omega T$ , continuously increasing  $E$  while the string remains taut.

## 2.4 Numerical method

Because Eq. (9) is a second-order nonlinear differential equation, it is converted into a first-order state-space system suitable for integration with `ode45`. The cumulative external work performed by the spool torque,  $W(t) = \int_0^t a\omega T dt$ , is incorporated as a third state variable according to Eq. (11):

$$\mathbf{X} = \begin{pmatrix} \phi \\ \dot{\phi} \\ W \end{pmatrix}, \quad \frac{d\mathbf{X}}{dt} = \begin{pmatrix} \dot{\phi} \\ \ddot{\phi} \text{ from Eq. (9)} \\ a\omega T \end{pmatrix}, \quad \mathbf{X}(0) = \begin{pmatrix} 0 \\ \dot{x}_P(0)/\xi_0 \\ 0 \end{pmatrix}. \quad (12)$$

This system is implemented in the MATLAB function `diffeq`. Numerical tolerances are set to  $10^{-8}$  (relative) and  $10^{-10}$  (absolute) to ensure accurate mechanical energy conservation in the unforced cases ( $\omega = 0$ ). An event function, `stopfun`, monitors terminal conditions: integration halts if  $\xi = 0$  (impact of the particle on the spool) or if  $T = 0$  (loss of string tension). The nominal parameters are  $g = 9.81 \text{ m/s}^2$ ,  $m = 0.1 \text{ kg}$ ,  $a = 0.2 \text{ m}$ ,  $\xi_0 = 1 \text{ m}$ , and  $\phi(0) = 0$ . The simulated cases and termination conditions are listed in Table 1.

Table 1: The five cases studied and how each integration ends.

| Case | $\omega$ (rad/s) | $\dot{x}_P(0)$ (m/s) | $\dot{\phi}(0)$ (rad/s) | $t_{\text{end}}$ (s) | End of the integration     |
|------|------------------|----------------------|-------------------------|----------------------|----------------------------|
| 1    | 0.1              | 0                    | 0                       | 50.00                | $\xi = 0$ (hits the spool) |
| 2    | 0                | -1                   | -1                      | 20                   | no event (oscillation)     |
| 3    | 0                | -10                  | -10                     | 0.2709               | $\xi = 0$ (hits the spool) |
| 4    | 0                | 4                    | 4                       | 1.693                | $T = 0$ (string loose)     |
| 5    | 0.1              | 1                    | 1                       | 35.51                | $\xi = 0$ (hits the spool) |

### 3 Results

Figure 2 shows  $\phi$ ,  $\xi$  and  $T$  for each case, Fig. 3 the trajectories and Fig. 4 the energy.

#### 3.1 Case 1: rotating spool, particle starts at rest

In Case 1, the particle is released from rest in the vertical downward position ( $\phi(0) = 0$ ,  $\dot{\phi}(0) = 0$ ). In the absence of lateral excitation, no angular oscillation is initiated, and the angle remains identically zero ( $\phi(t) = 0$ ). The spool rotates at constant rate, winding the string at the linear speed  $a\omega = 0.02 \text{ m/s}$ . Consequently, the unwound length decreases linearly as  $\xi(t) = 1 - 0.02t$  (Fig. 2, top row). Because the vertical ascent occurs at constant velocity, the acceleration is zero, and the string tension equals the gravitational weight:  $T = mg = 0.981 \text{ N}$ . At  $t = \xi_0/(a\omega) = 50 \text{ s}$ , the entire 1-meter string is wound, and the particle contacts the spool surface. Because the particle velocity matches the tangential velocity of the spool at contact, no impulsive impact occurs. The mechanical energy increases linearly at the rate  $a\omega mg = 19.6 \text{ mW}$  (Fig. 4a), with a total energy gain of  $mg\xi_0 = 0.981 \text{ J}$  corresponding to the gravitational potential energy gained over 1 m.

#### 3.2 Case 2: fixed spool, small initial velocity

In Case 2, the spool remains stationary ( $\omega = 0$ ), and an initial horizontal velocity of  $-1 \text{ m/s}$  is applied to the left. The particle exhibits pendulum-like oscillations with variable string length. The motion displays geometric asymmetry: when the particle swings to the left ( $\phi < 0$ ), string wraps onto the spool cylinder and the effective length decreases to a minimum of  $0.934 \text{ m}$ ; when it swings to the right ( $\phi > 0$ ), string unwinds and extends to  $1.063 \text{ m}$ . The angular range extends between  $-18.8^\circ$  and  $+18.0^\circ$ . The oscillation period is  $2.018 \text{ s}$ , slightly longer than the  $2.006 \text{ s}$  period of a standard 1-meter simple pendulum. The tension reaches a maximum of  $T = 1.08 \text{ N}$  at the lowest point where the velocity peaks, and decreases to  $0.93 \text{ N}$  at the turning points where  $\dot{\phi} = 0$ . The tension remains strictly positive, preventing string slackening throughout the 20-second simulation.

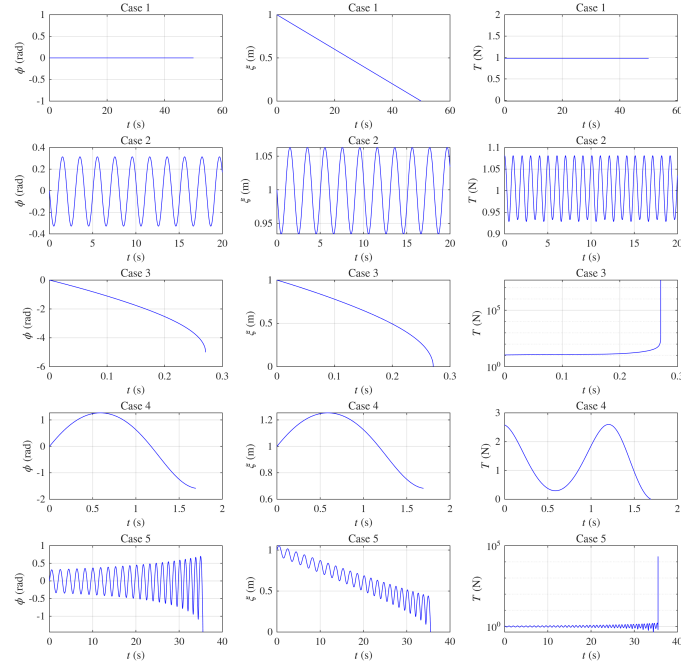


Figure 2: Angle  $\phi$ , free length  $\xi$  and tension  $T$  against time for the five cases (one case per row). In Cases 3 and 5 the tension is plotted in logarithmic scale because it becomes very large at the end.

### 3.3 Case 3: fixed spool, large initial velocity

In Case 3, a large initial velocity of  $-10$  m/s is applied to the left with  $\omega = 0$ . The particle wraps around the spool in a rapid inward spiral. The unwound length reaches zero at  $\phi = -\xi_0/a = -5$  rad (approximately 0.8 turns). Contact occurs at  $t_f = 0.2709$  s at coordinates (0.057, 0.192) m on the upper surface of the spool (Fig. 3).

Near the contact point, the linear speed of the particle remains bounded at  $v_s = 8.75$  m/s by energy conservation. However, because the instantaneous radius of curvature  $\xi$  approaches zero, the angular velocity must diverge to preserve linear velocity:  $|\dot{\phi}| \approx v_s/\xi$ . The required centripetal force causes the string tension to escalate rapidly according to:

$$\dot{\phi} \simeq -\sqrt{\frac{v_s}{2a(t_f - t)}}, \quad T \simeq \frac{mv_s^2}{\xi}. \quad (13)$$

Both  $\dot{\phi}$  and  $T$  exhibit power-law divergence proportional to  $(t_f - t)^{-1/2}$ , confirmed by the  $-1/2$  slope on the logarithmic scale in Figure 5a. At  $t = t_f$ , the geometric singularity terminates the validity of the involute model upon physical impact.

### 3.4 Case 4: fixed spool, loss of tension

In Case 4, an initial positive velocity of 4 m/s is applied to the right with  $\omega = 0$ . The particle ascends to  $\phi = 72.6^\circ$  (1.267 rad) at  $t = 0.588$  s, where it halts and reverses direction. Accelerating through the bottom position, it climbs along the left side, wrapping onto the spool. Once the angle passes horizontal ( $\phi < -90^\circ$ ), the unit vector  $\mathbf{e}_t$  points upward, meaning the gravitational component along the string acts inward toward the spool center. To maintain tension, the centripetal acceleration must satisfy  $\xi\dot{\phi}^2 \geq -g \cos \phi$ . At  $t = 1.693$  s, the linear speed drops to 0.318 m/s, and the tension vanishes at  $\phi = -90.9^\circ$  and  $\xi = 0.683$  m (Fig. 2, fourth row). The string becomes slack, violating the tensile constraint. Beyond this point, the particle transitions into unconstrained ballistic flight under gravity. Mechanical energy is strictly conserved at  $E = -0.181$  J up to the slackening event.

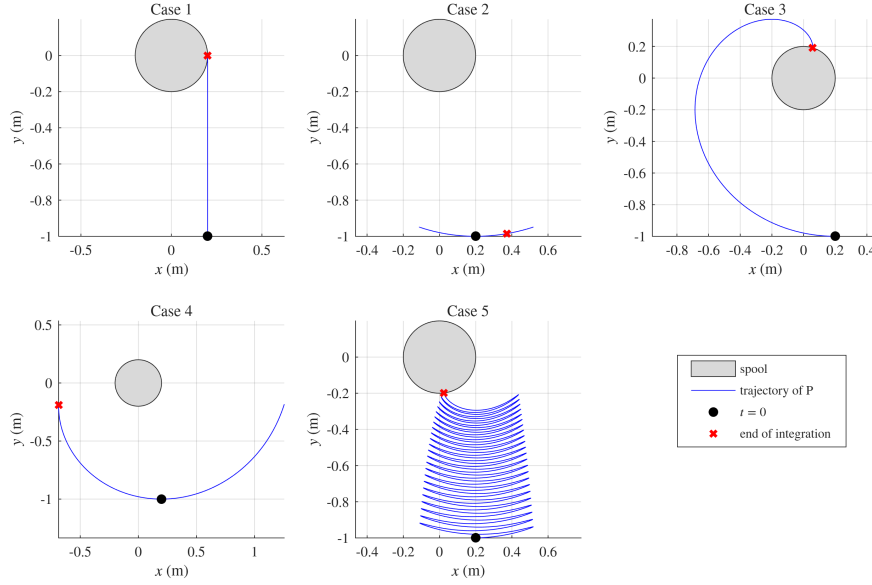


Figure 3: Trajectory of  $P$  in the  $Oxy$  plane for each case. The grey circle is the spool (drawn at  $t = 0$ ). The black dot is the start and the red cross the end of the integration.

### 3.5 Case 5: rotating spool with initial velocity

In Case 5, the spool rotates at  $\omega = 0.1$  rad/s while an initial velocity of 1 m/s is applied to the particle. On average, the unwound length decreases at the linear rate of  $a\omega = 0.02$  m/s. As the string shortens, the oscillation frequency increases, reducing the period from 2.0 s initially to 1.15 s near termination. Concurrently, the oscillation amplitude grows from  $\pm 0.3$  rad to  $[-1.45, 0.70]$  rad because the rotating spool performs net positive work on the particle during each cycle. The tension remains positive throughout (minimum 0.46 N), preventing slackening. At  $t = 35.51$  s, the unwound length vanishes ( $\xi = 0$ ), and the particle impacts the lower surface of the spool at  $(0.024, -0.199)$  m (Fig. 3), accompanied by asymptotic escalation of  $\dot{\phi}$  and  $T$  as in Case 3.

### 3.6 Energy

The mechanical energy histories provide clear validation of the governing dynamics. For all unforced cases with  $\omega = 0$  (Cases 2, 3, and 4), the string tension is everywhere orthogonal to the velocity vector, performing zero work. Mechanical energy is rigorously conserved (Fig. 4b–d). For Cases 2 and 4, the relative numerical drift remains below  $4 \times 10^{-8}$  (down to  $3.6 \times 10^{-9}$  in Case 2, Fig. 4f). In Case 3, the relative error stays below  $10^{-9}$  across almost the entire trajectory, rising to  $1.8 \times 10^{-6}$  ( $|\Delta E| \approx 2.9 \times 10^{-6}$  J) only at the final integration step because of the singular acceleration near impact. When the spool rotates ( $\omega = 0.1$  rad/s), the boundary condition performs work on the system at the rate  $dE/dt = a\omega T$ . In Case 1, the constant tension produces linear energy growth. In Case 5, tension oscillations induce cyclic fluctuations superimposed on an overall energy gain of 0.74 J. Simultaneous integration of the external work  $W(t) = \int_0^t a\omega T dt$  with `ode45` demonstrates exact agreement between  $E(t) - E_0$  and  $W(t)$  (Fig. 5b), with residuals below  $10^{-8}$  J.

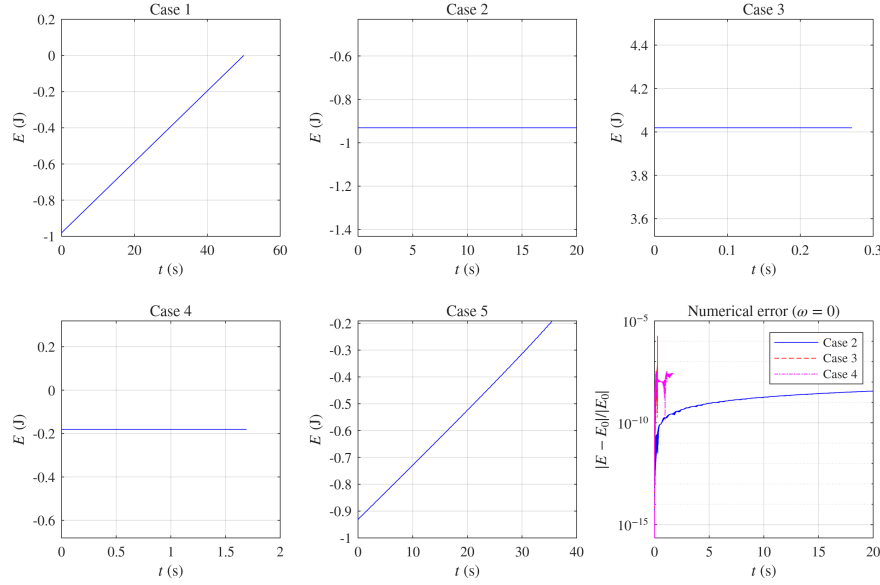


Figure 4: (a–e) Mechanical energy  $E$  against time in Cases 1–5. For  $\omega = 0$  the vertical axis covers  $\pm 0.5$  J around  $E_0$ . (f) Relative change of the energy for the cases with  $\omega = 0$ .

#### 4 Discussion: motion after the string becomes loose

In Case 4 the integration stops when  $T = 0$ . After that moment the string is loose, so Eq. (9) is not valid anymore: the particle is free and moves only under gravity, with two degrees of freedom. To continue the simulation, the code could be changed like this (always using `ode45`):

1. Take the values at the event ( $t^*$ ,  $\phi^*$ ,  $\dot{\phi}^*$ ) and calculate the position and velocity of  $P$  in  $x$  and  $y$  with Eqs. (3) and (4). The new state is  $\mathbf{X}_2 = [x, y, \dot{x}, \dot{y}]^T$ .
2. Write the free fall,  $\ddot{x} = 0$ ,  $\ddot{y} = -g$ , as a first order system for  $\mathbf{X}_2$  and integrate it with `ode45` from  $t^*$ .
3. Use a new event function with two events: (i) the string becomes tight again, which happens when the wrapped part plus the straight part is equal to  $l$ ; (ii) the particle hits the spool,  $\sqrt{x^2 + y^2} = a$ .
4. When the string becomes taut again, an impulsive normal constraint force acts on the particle. The velocity component along the string is instantaneously set to  $-a\omega$  (Eq. (4)), while the transverse component is preserved. This resembles an inelastic impact, dissipating a small portion of kinetic energy. From the updated Cartesian velocity, the polar state  $(\phi, \dot{\phi})$  is re-initialized to resume integration with Eq. (9) via `ode45`.
5. The overall algorithm alternates between taut-string and free-flight phases within a hybrid simulation loop until impact occurs or the time limit is reached, concatenating all solution pieces for visualization.

In Case 4 the speed at the event is only 0.32 m/s and almost vertical, so the particle would make a short free flight and then the string would become tight again, with a small loss of energy.

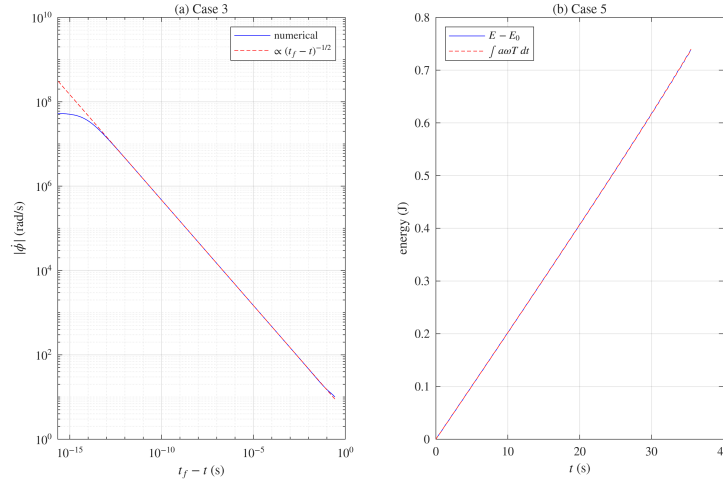


Figure 5: (a) Angular velocity of Case 3 as a function of the time remaining to the collision  $t_f - t$ , in logarithmic scale, together with the asymptotic law. (b) Work–energy check in Case 5: increment of mechanical energy and integral of the power  $a\omega T$ .

## 5 Conclusions

The dynamics of a particle constrained by a winding string around a rotating cylinder was analyzed and solved numerically using `ode45`. Despite the continuous geometric winding, the system possesses a single degree of freedom parametrized by the angular position  $\phi$ , behaving as a generalized pendulum with time- and state-dependent length.

The mechanical energy balance governs the diverse responses observed across all five cases. When the spool is stationary ( $\omega = 0$ ), the constraint forces perform no work, and mechanical energy is conserved with high numerical accuracy (relative drift below  $4 \times 10^{-8}$  in Cases 2 and 4, and below  $2 \times 10^{-6}$  at the singularity in Case 3). Depending on the initial velocity, the motion manifests as bounded periodic oscillation (Case 2), rapid spiral winding leading to spool impact (Case 3), or string slackening upon deceleration above the spool (Case 4). Conversely, when the spool rotates ( $\omega > 0$ ), work is continuously injected into the system at the rate  $a\omega T$ , lifting the particle at constant velocity (Case 1) or progressively amplifying the oscillation amplitude until spool impact occurs (Case 5). Numerical evaluation of the work integral using `ode45` verified the work–energy relation with high precision.

Finally, the operational boundaries of the single-degree-of-freedom formulation were identified. The mathematical model terminates under two distinct physical events: spool collision ( $\xi \rightarrow 0$ ), where angular velocity and tension exhibit power-law divergence  $(t_f - t)^{-1/2}$ , and loss of tension ( $T = 0$ ), where the string ceases to support compressive forces. Beyond the slackening threshold, the particle uncouples from the spool constraint, requiring a two-degree-of-freedom ballistic model in Cartesian coordinates until tension is impulsively restored.

Therefore, the experiment successfully demonstrated the application of numerical integration with `ode45` to solve the equations of motion for a particle connected to a spool. The results provide clear insight into the dynamics under varying initial conditions and angular velocities. Energy conservation was verified in cases where the string maintained tension without rotation, while the event function correctly stopped the simulation when physical thresholds were reached. Future extensions would focus on implementing the piecewise ballistic flight algorithm for the unconstrained phases following string slackening.

## References

- [1] Aerospace Engineering Department (UC3M), *Mechanics Applied to Aerospace Engineering: Laboratory Guidelines*, Version 2026.
- [2] The MathWorks, Inc., *MATLAB Documentation: ode45, odeset*.